Asmi Umesh Pulgam

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Professional Summary

Software Engineer pursuing an M.S. in Computer Science at Arizona State University with experience in building scalable full-stack web applications, cloud-native microservices, and AI-powered developer tools. Proficient in React.js, Python, REST APIs, and modern frontend development with experience designing responsive user interfaces and integrating complex backend systems. Passionate about building intuitive data-driven web experiences.

Education

Arizona State University — Tempe, AZ, US
Master of Science in Computer Science; GPA: 4.00 / 4.00 — *Aug 2025 – May 2027*

Savitribai Phule Pune University — Pune, MH, India
Bachelor of Engineering in Computer Engineering, Honors in Data Science; GPA: 3.50 / 4.00 — *Aug 2021 – May 2025*

Skills

Programming Languages: Python, Java, JavaScript, TypeScript
Frontend: React.js, Next.js, HTML5, CSS3, Bootstrap
AI/ML: TensorFlow, PyTorch, LLM, LangChain, RAG, NumPy, Pandas, Scikit-Learn, Matplotlib, Agentic AI, Fine-Tuning
Databases: MySQL, PostgreSQL, MongoDB, Redis, PySpark, Kafka, Pinecone, Snowflake
Cloud Platforms: AWS (EC2, S3, Lambda, DynamoDB, SQS, ELB, KeyCloak), Docker, Kubernetes, CI/CD
Tools: Git, GitHub, Claude API, Gemini API, Vercel, VS Code, REST API, Plaid API, Prompt Engineering

Work Experience

Software Engineer Intern — Feb 2024 – Aug 2025
Infosys — *Pune, India*

- Transformed a monolithic banking platform into reusable cloud-native REST microservices by migrating 15+ SOAP services with Java, Spring Boot, MongoDB, and AWS, improving API response times by 35% while reducing maintenance effort by 50%.
- Automated CI/CD and containerized deployment pipelines using Jenkins, GitLab, Docker, and Kubernetes, achieving 85% deployment automation, 60% faster releases, and 30% lower AWS infrastructure costs.

Projects

Natural Language-to-SQL Query Engine | https://github.com/asmipulgam/fieldcode

- ***Python, PostgreSQL, Anthropic Claude API, SQL, LangChain, Git, MCP, Redis, Snowflake, HTML5, CSS3***
- Engineered a full-stack responsive AI platform enabling 100+ non-technical users to query PostgreSQL, MySQL, Redis, and Snowflake databases using natural language through an intuitive web interface.
- Built scalable backend services using Python, integrating Anthropic Claude API, LangChain, and secure MCP-based connectors to translate natural language into optimized SQL queries across multiple data sources.
- Implemented LangChain query caching, optimized backend performance, and developed PyTest-based unit tests to validate query generation and API functionality, reducing average query response time by 40% while improving application reliability.

Agentic AI Financial Advisor Platform with RAG | sparkyfi.vercel.app

- ***Next.js, TypeScript, Gemini 2.5 Flash, Pinecone, Plaid API, CSS, Vercel, RAG, Agentic AI***
- Developed a full-stack AI financial advisor leveraging Gemini 2.5 Flash, Plaid API, and ElevenLabs, enabling users to analyze bank statements, insurance policies, leases, and loan documents through conversational AI and voice interaction.
- Architected a production-grade RAG and agentic AI pipeline with Pinecone vector search and semantic retrieval across 2,800+ State Farm and CFPB records, delivering personalized, context-aware financial recommendations.

Distributed Database Performance Optimization System | github.com/asmipulgam/Geo-Distributed-HealthCare-Database-System

- ***CockroachDB, MongoDB, Docker, SQL, Python, Distributed Systems***
- Designed a cross-region database platform using CockroachDB, MongoDB, Docker, and SQL, providing fault-tolerant, highly available data storage across multi-region clusters.
- Optimized query performance by refining replication, data placement, and consistency models, reducing cross-region latency by 30% and improving throughput by 25%.

Privacy-Preserving Language Model Fine-Tuning and Deployment | GitHub

- ***Python, PyTorch, Hugging Face Transformers, TinyLlama, Qwen2.5, LoRA, QLoRA***
- Built an on-device hate speech detection system using TinyLlama, Qwen2.5, PyTorch, and Hugging Face Transformers, ensuring 100% on-device inference without transmitting sensitive user data.
- Enhanced Small Language Model training efficiency using LoRA and QLoRA, achieving 2-hour fine-tuning on a 4GB GPU and increasing offensive-content recall from 39% to 55%.

Leadership Experience

Technical Lead — Aug 2023 – May 2024
Google Developer Student Club (GDSC)

- Led technical workshops and coordinated hackathons and speaker sessions, guiding students through web development, cloud services, application of data structures and algorithms and AI/ML fundamentals.